

Qualification of the OpenMC Radiation Transport Code for ITER

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1 Abstract

OpenMC is an open-source Monte Carlo particle transport code used for neutron and photon transport simulations. The code is developed by an international community of researchers, engineers, and contributors from national laboratories, universities, industry, and research institutions. Development is coordinated through a public GitHub repository, allowing transparent review, validation, and continuous improvement of the code. OpenMC supports continuous-energy nuclear data, constructive solid geometry (CSG), CAD-based geometries through DAGMC, and structured and unstructured mesh tallies. The code is designed with a focus on accuracy, scalability, and a Python-based workflow that simplifies model generation, execution, and post-processing.

OpenMC is applied to a wide range of problems in reactor physics, radiation shielding, activation analysis, criticality safety, isotope production, and fusion neutronics. It provides capabilities for eigenvalue and fixed-source calculations, depletion and activation analyses, variance reduction, and coupling to external multiphysics workflows. Parallel execution using MPI and OpenMP enables simulations ranging from small benchmark problems to large engineering models.

The Python API provides direct access to modelling, execution, and analysis tools, allowing reproducible simulation workflows and integration with external software. Ongoing development continues to expand transport, depletion, and data-processing capabilities while maintaining an open development model. As a result, OpenMC has become a widely used platform for research, code development, verification and validation studies, and engineering applications requiring high-fidelity particle transport calculations.

This document describes the verification and validation (V&V) of the OpenMC Monte Carlo radiation transport code in accordance with *ITER_D_VP6DJI v2.1 – Instructions on qualification of radiation transport codes*.

The objective is to demonstrate that OpenMC is suitable for use in *ITER radiation transport calculations for non-PIA activities, or PIAs which are not part of the safety demonstration*.

The initial version of this qualification document was generated using OpenMC v0.15.3. Subsequent releases will be automatically tested and the results refreshed. New functionality will be appended to the report.

Code name	OpenMC
Version	v0.15.3
Git commit	27e38e894697bb32a1dac7848d2618818b6b8daf
Execution mode	Fixed source Monte Carlo
Particles	Neutrons, photons
Nuclear data	ENDF/B-VIII.0

2 Installation and User Manual

The OpenMC documentation, theory descriptions, and user manuals are publicly available and provide a complete description of the implemented numerical methods at: OpenMC Documentation. This report includes a small subset of the documentation for reference purposes and is available in Appendix B.

3 Verification and Validation

All test executions and results reported herein are fully automated and reproducible. The results and input decks are hosted by:

- JADE project
- OpenMC Fundamental Testing git repository

The raw results generated by JADE can be retrieved from JADE Raw Results or visualized online using the WebApp.

3.1 Results Summary

Table 2: Results for Functional Tests

Requirement ID	Description	Status
7.2.1.1	Neutron/Photon flux	PASS
7.2.1.2.1	Nuclear heat	PASS
7.2.1.2.2	Tritium Production Rate	PASS
7.2.1.2.3	Dose	PASS
7.2.1.2.4	Helium production rate	PASS
7.2.1.2.5	DPA	PASS

Table 3: Results for Fundamental Tests

Requirement ID	Description	Status
7.2.2.1	Test of statistical error	PASS
7.2.2.2	Test of statistical error with weight window	PASS
7.2.2.3	Test of random number generator - Knuth	PASS
7.2.2.3	Test of random number generator - Diehard	PASS
7.2.2.4	Test of source sampling Case 1	PASS
7.2.2.4	Test of source sampling Case 2	PASS
7.2.2.4	Test of source sampling Case 3	PASS
7.2.2.4	Test of source sampling Case 4	PASS
7.2.2.5	Test of conservation of energy for nuclear reaction	PASS

Detailed numerical results, plots, and statistical tests are provided in the Appendix A and archived as objective evidence in the above mentioned GitHub repositories.

4 Conclusion

Based on the verification and validation activities documented in this report, **OpenMC version v0.15.3** is in our opinion considered suitable for use in ITER radiation transport calculations within the scope defined in ITER_D_VP6DJL v2.1.

A Detailed results

A.1 Functional tests

The main purpose of the functional testing is to ensure the correctness of the transport calculation functions of the code.

A.1.1 7.2.1.1 Neutron/Photon flux

Test case: Calculate the spectra distribution in three locations in ITER tokamak (e.g. equatorial inboard and outboard blankets, divertor region including TFC leg) as specified in reference. The energy bins for comparing spectra are reported in Tab.1. The calculation should be performed based on the ITER C-lite model or reference models released after C-lite.

Test method: The code should provide geometry visualization method using the ray tracing method as same as that used in the nuclear transport simulation. First compare the geometry visualized by the code with the anticipated geometry, if it passes the comparison, it would verify the correctness of the complex geometry used for the transport simulation of the code. Then compare the calculation results with those calculated by the reference code (such as MCNP), accept the calculation results as correct if they conform the acceptance criterial described in section 7.3.

A.1.2 7.2.1.2.1 Nuclear heat

Test case: Calculate the nuclear heat deposition on a number of specific locations including the first wall (FW), blanket (BLK), vacuum vessel (VV), TF coil (TFC) and divertor for a number of the specific materials separately for neutron and photon in the specific energy bin structures. Table 2 shows the materials, location and energy bin structure for this calculation. The calculation should be performed based on the ITER C-lite model or reference models released after C-lite.

Test method: Compare the calculation results with those calculated by the reference code (such as MCNP), accept the calculation results as correct if they conform the acceptance criterial described in section 7.3.

A.1.3 7.2.1.2.2 Tritium Production Rate

Test case: HCPB.

Test method: Compare the calculation results with experiment measurements (The positions are POS 4,6,8 in HCPB model), accept the calculation results as correct if they conform the acceptance criterial described in section 7.3.

A.1.4 7.2.1.2.3 Dose

Test case: FNG/dose.

Test method: Compare the calculation results with experiment measurements (The decay time are 15,30,60 days) or the results calculated by the reference code (such as MCNP), accept the calculation results as correct if they conform the acceptance criterial described in section 7.3.

A.1.5 7.2.1.2.4 Helium production rate

Test case: FNG-BLKT.

Test method: Compare the calculation results with experiment measurements or the results calculated by the reference code (such as MCNP), accept the calculation results as correct if they conform the acceptance criterial described in section 7.3.

A.1.6 7.2.1.2.5 DPA

Test case: DT-FW. *Model could not be provided by the IO or retrieved from anywhere. The IO agreed C-model can be used.*

Test method: Compare the calculation results with those calculated by the reference code (such as MCNP), accept the calculation results as correct if they conform the acceptance criterial described in section 7.3.

A.2 Fundamental tests

The main purpose of this testing is to verify the implementations of the Monte Carlo method and the preservation laws.

A.2.1 7.2.2.1 Test of statistical error

Purpose: To verify the trend and correctness of statistical error.

Test case: The model is a simple vacuum sphere with radius 5 cm, and the isotropic neutron source is defined in the position 10 cm from the centre of the sphere with mono energy 14MeV. The neutron flux in the sphere is counted for different particle histories.

Test method: Perform three independent calculations with increasing 10,000, 100,000 and 1,000,000 particle histories. Compare the trend of the statistical error, check if the values decrease linearly with the square root of the history number. And then compare the statistical errors of neutron flux in the vacuum sphere with the theoretical value as *TEXT MISSING IN ORIGINAL DOCUMENT. From context, the continuation of the sentence was assume to be: the inverse square root dependence on the number of particle histories.*

Problem was setup as described and ran with OpenMC. Results:

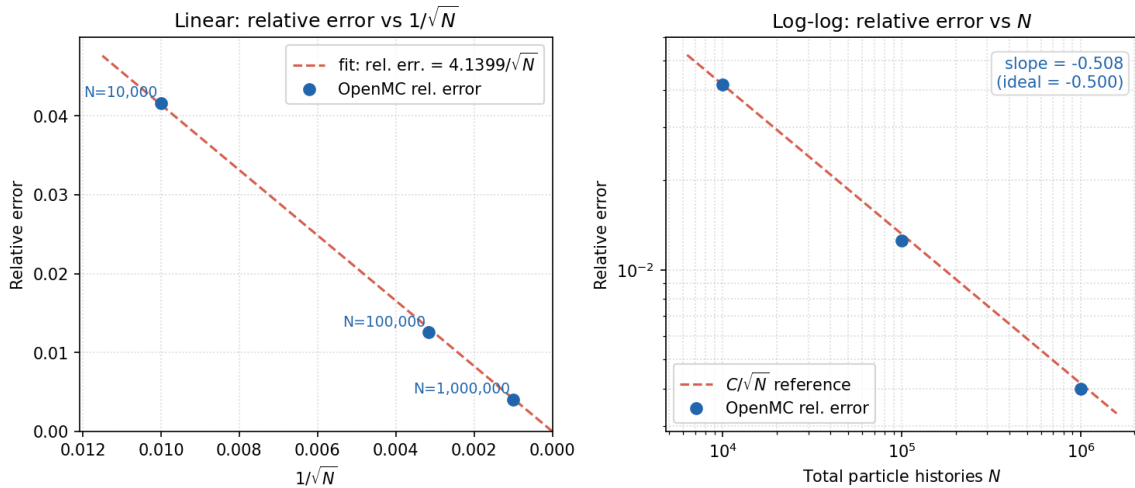


Figure 1: Statistical convergence of results as a function of increasing simulated particle histories.

A.2.2 7.2.2.2 Test of statistical error with weight window

Purpose: To verify the trend and correctness of statistical error with/without weight window variance reduction techniques.

Test case: The models of the variance reduction examples. Test method: Perform several independent calculations with increasing particle histories 10,000, 100,000 and 1,000,000 with and without use of the variance reduction technique(s). Then compare the trend of the statistical errors, check whether the statistical error with variance reduction on is smaller than that without use of the VR technique at the same histories. When calculating with a weight window, the

variance of the variance should be calculated, to verify the convergence of the calculation results. Only results with the expected behaviour of the mean, variance, and the variance of the sample variance should be considered valid for the statistical error test.

Problem was setup as described in and ran with OpenMC. The model is a simple concrete cube (2 m sides) with a 14 MeV neutron source. The quantities of interest are tallies in cells at increasing shield thicknesses in 10 cm increments. The Random-Ray solver is used to create FW-CADIS based weight windows on the fly. Results:

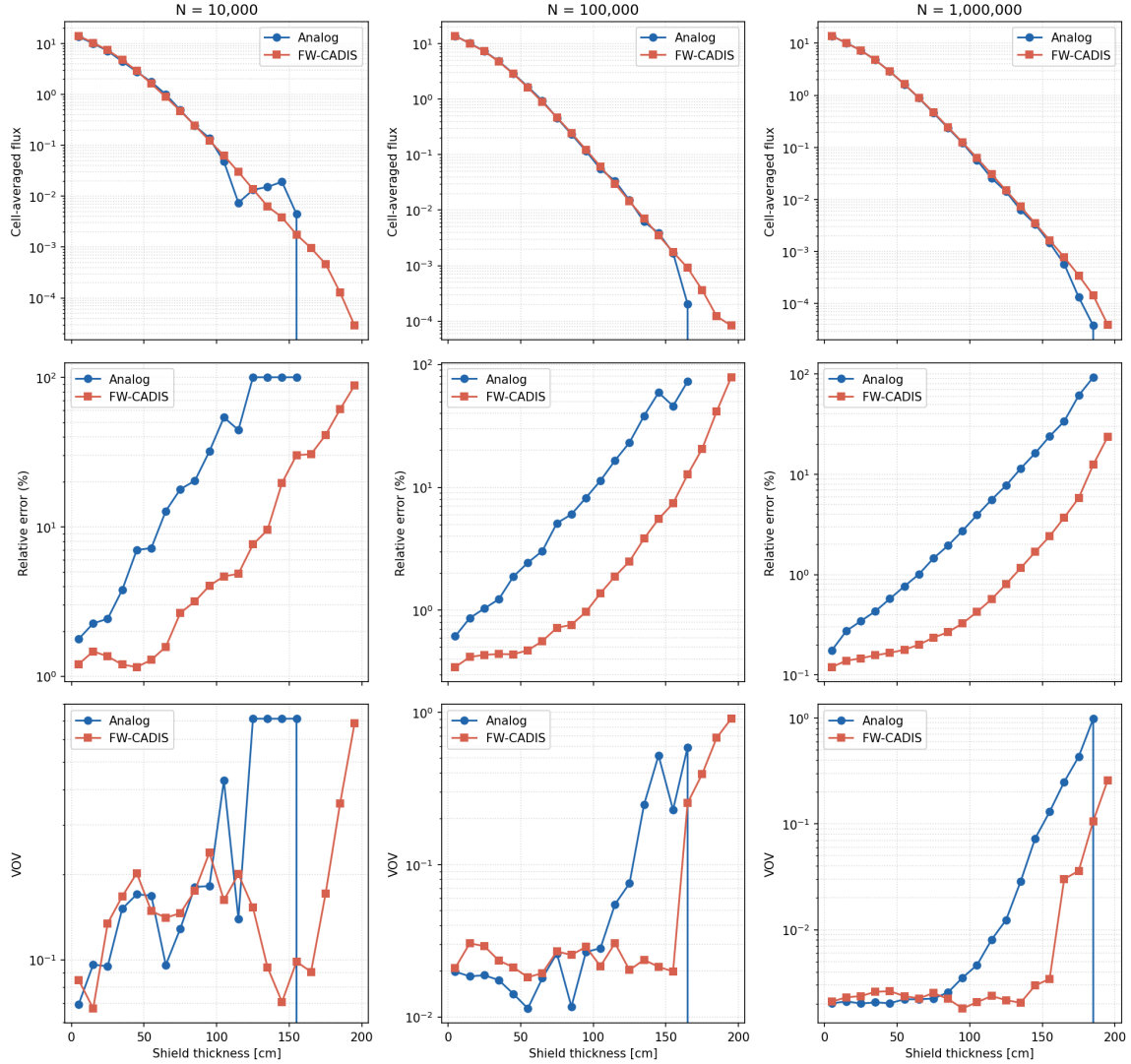


Figure 2: Comparison of analog and weight window accelerated simulation result as a function of shield thickness. Each column represent a 10-fold increase in simulated particle histories. Top row mean results, middle row relative error, bottom row Variance of Variance.

A.2.3 7.2.2.3 Test of random number generator

Purpose: To verify the correctness of the random number generator. As random number generators in Monte Carlo codes are implemented based on LCGs, this test is aiming to verify the correctness of the LCGs implemented in the code.

Test case 1: The standard statistical tests for random number generators described by Knuth or variations on Knuth's statistical tests.

The problem was setup as described. The random number generator was accessed via the DLL. A series of tests were run including: Frequency, Serial, Runs, Gap, Poker, Autocorrelation.

For more details you can look at the full test report in: `/fundamental_testing/7223_rng/1` of the OpenMC Fundamental Testing git repository. Test case 2: Marsaglia's DIEHARD test suite for random number generators.

Test Method: Write random stream generating program based on the LCGs code implemented in the code. Apply the two test cases to the random stream generated by the random stream generating program, if it passes the acceptance criteria, the random number generator of the code can be considered as acceptable.

The problem was setup as described. The random number generator was accessed via the DLL. The Diehard test suite was run using the the modern implementation as part of Dieharder. Dieharder can be installed from the Dieharder git repository or in Ubuntu/Debian simply by `sudo apt-get install dieharder`.

List of tests performed:

- Birthday Spacings Test – `-d 0` (Diehard Birthdays Test)
- Binary Rank 31x31 and 32x32 – `-d 2` (Diehard 32x32 Binary Rank Test)
- Binary Rank 6x8 – `-d 3` (Diehard 6x8 Binary Rank Test)
- Bitstream – `-d 4` (Diehard Bitstream Test)
- OPSO (Overlapping-Pairs-Sparse-Occupancy) – `-d 5` (Diehard OPSO Test)
- OQSO (Overlapping-Quadruples-Sparse-Occupancy) – `-d 6` (Diehard OQSO Test)
- DNA – `-d 7` (Diehard DNA Test)
- Count the 1s (stream of bits) – `-d 8` (Diehard Count the 1s Stream Test)
- Count the 1s (specific byte) – `-d 9` (Diehard Count the 1s Byte Test)
- Parking Lot Test – `-d 10` (Diehard Parking Lot Test)
- Minimum Distance (2D circles/spheres) – `-d 11` (Diehard Minimum Distance Test)
- 3D Spheres (Minimum Distance) – `-d 12` (Diehard 3D Sphere Test)
- Squeeze Test – `-d 13` (Diehard Squeeze Test)
- Runs Test (Wald-Wolfowitz) – `-d 15` (Diehard Runs Test)
- Craps Test – `-d 16` (Diehard Craps Test)

For more details you can look at the full test report in: `/fundamental_testing/7223_rng/2` of the OpenMC Fundamental Testing git repository.

A.2.4 7.2.2.4 Test of source sampling

Purpose: To verify the correctness of the source sampling with probability distribution.

Test case 1: The model is a simple vacuum sphere with radius 5 cm, and the isotropic neutron source is defined in the centre of the sphere with specific energy distribution. Tally the neutron energy integrated over the sphere surface with refined energy bins.

Test method 1: Perform calculation until the tally results in each energy bin converge. Compare the calculation results with the specific energy distribution of the source definition, consider the code passes this test if the results meet the requirements described in section 7.3.

Problem was setup as described and ran with OpenMC. A triangular energy distribution was chosen. Results:

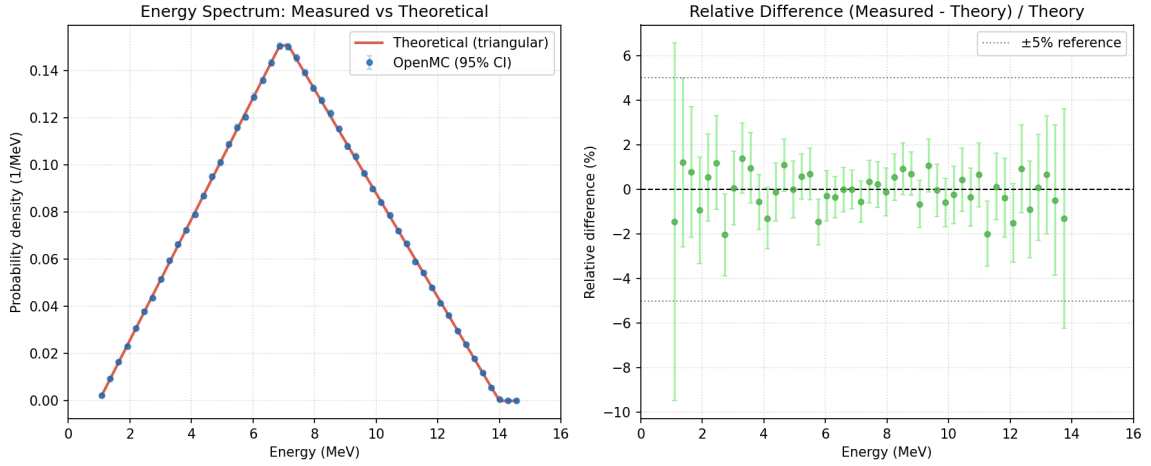


Figure 3: Simulated vs theoretical source energy distribution.

Test case 2: The model is a simple vacuum cuboid pipe with length 10 cm*0.1 cm*0.1 cm (the coordinate of centre point is zero, dimension is defined in the order of x, y, z axis). The neutron source is defined inside the plate with single direction along the z axis, and the distribution of the source position is sampled by a specific distribution along x axis and constant along y and z axis. The surface current of the top surface (perpendicular to z axis and on the top of the cuboid pipe) segmented evenly to 100 segments by refined surfaces perpendicular to x axis are counted.

Test method 2: Perform calculations until all the tallies in all the segments converge. Compare the results to the source position sampling, the code would be considered passed this test if the results meet the criterial described in section 7.3.

Problem was setup as described and ran with OpenMC. A triangular positional distibtuion was chosen. Results:

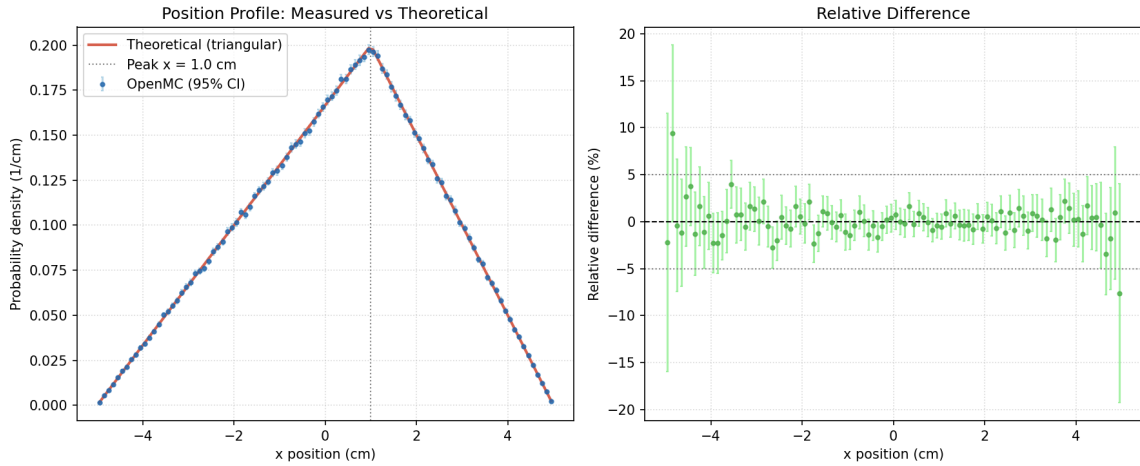


Figure 4: Simulated vs theoretical source positional distribution.

Test case 3: The model is a simple vacuum circular slab with 5 cm radius and 1 cm thickness along with z axis, and the mono neutron source (14MeV) is defined in the centre line of the slab with specific direction distribution (perpendicular to z axis with refined distributions). Split the slab into 365 segments that each segment is reference to 1 degree, and tally the neutron flux in each segment.

Test method 3: Perform calculations until all the tallies in all the source bins converge.

Compare the results to the source direction sampling, the code would be considered passed that this test if the results met the criteria described in section 7.3.

Problem was setup as described and ran with OpenMC. A sinusoidal non-uniform azimuthal distribution angular distribution was chosen. Results:

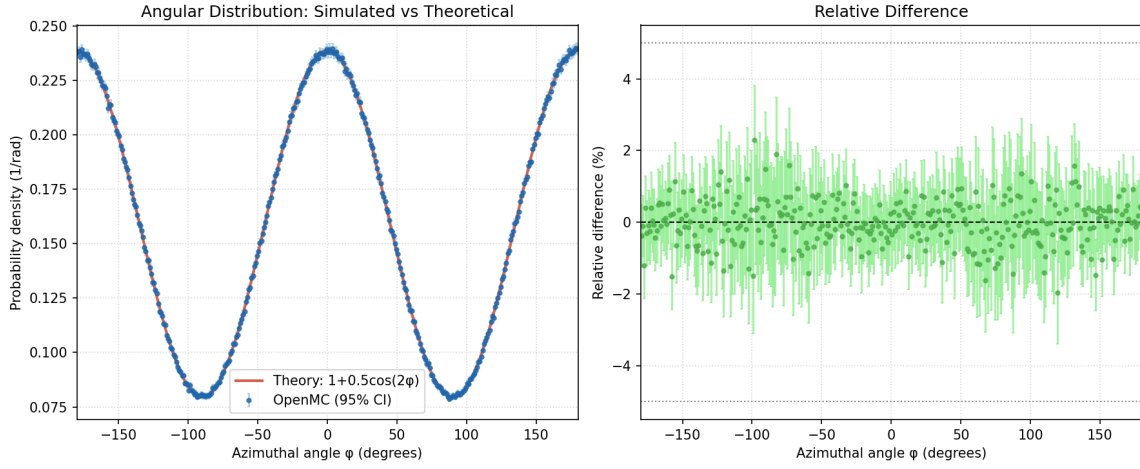


Figure 5: Simulated vs theoretical source angular distribution.

Test case 4 and the plasma solid with reflection surfaces is used as the test model. The neutron flux in the mesh with grid size of 10cm*10cm*10cm covering the C-lite model would be tallied.

Test method 4: Perform calculations until tallies of all the grids converge, compare the results with MCNP results, considered the code passed this test if the results meet the criteria described in section 7.3.

Problem was setup in MCNP and OpenMC according to the description. The ITER C-model plasma source was translated into OpenMC. Both the MCNP and OpenMC plasma source were positioned in to a voided wedge with reflective boundaries. Results:

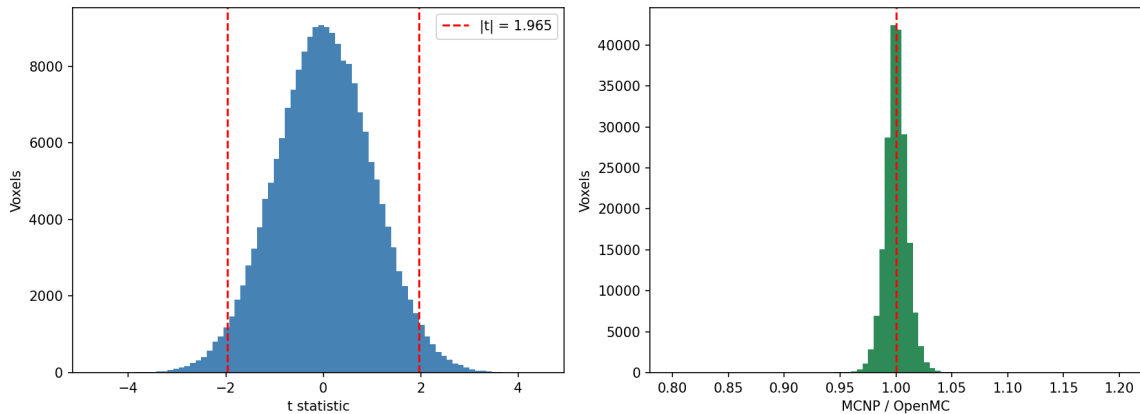


Figure 6: Student t-statistical test results and ratio of fluxes on a 10cm mesh in MCNP and OpenMC.

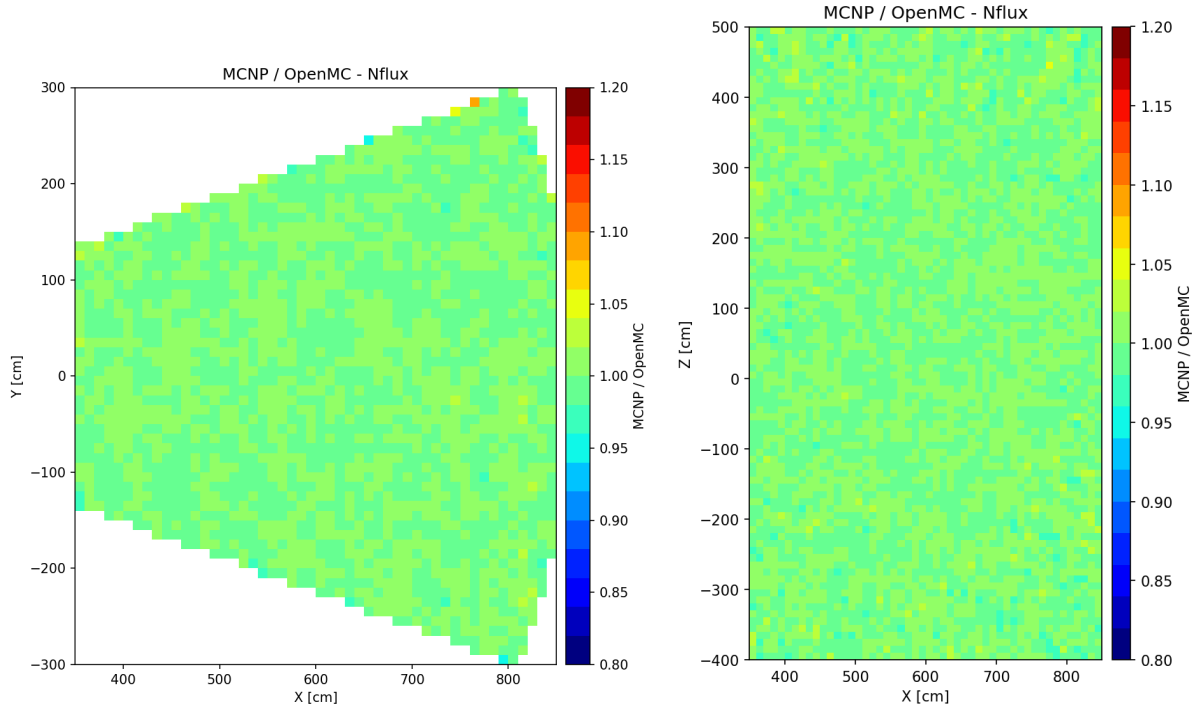


Figure 7: Ratio of MCNP and OpenMC neutron flux over plasma source with reflective boundaries. XY plane (left), XZ plane (right).

A.2.5 7.2.2.5 Test of conservation of energy for nuclear reaction

Purpose: To verify the correctness of energy conservation for nuclear reaction.

Test case: The model is a cuboid (50 cm*50 cm*50 cm) with nature iron. The neutron source is located in the bottom of the cuboid with direction to the negative half of z axis and mono energy 14MeV. The neutron and photon energy integrated over each surface of cuboid are counted, as well as heating inside the cuboid.

Test method: Perform the calculation with 100,000,000 particles, and count the entered and leakage neutron and photon energy independently.

Compare the entered neutron energy A, nuclear heat deposition of neutron and photon B and the leakage of neutron and photon energy C, then get the subtraction (B+C-A). The next step, the comparison should be done between the subtraction value and the (n,gamma) reaction deposition energy calculated by the code, consider the code passes this test if the reaction deposition energy has been included in the subtraction value. Problem was setup as described and ran with OpenMC. However the wording of the test is ambiguous. It is hard to prove quote: *if the reaction deposition energy has been included in the subtraction value* at the specified 14 MeV where several reaction (including thresholed) reactions occur that staurate the overall *subtraction value* as it is seen in Figure 8.

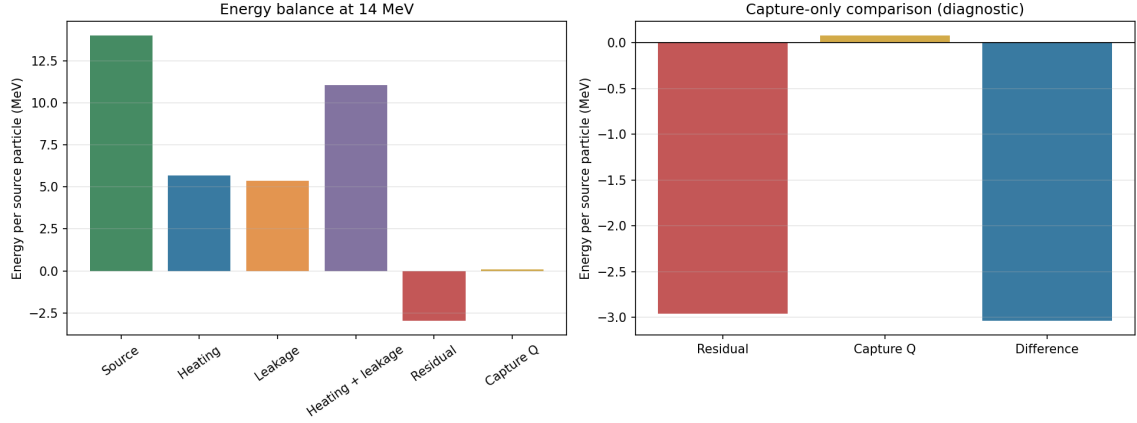


Figure 8: Energy balance with 14 MeV neutron source. A large residual energy is observed which can be attributed to several reaction channels such as: $(n,2n)$, (n,p) , (n,α)

We propose to ammend the test and run it with 1 MeV neutrons where the energy balance is much simpler and where indeed the subtraction value is close to the capture reaction deposition energy. Results with the 1 MeV source is shown in Figure 9. Both tests are performed, but the Summary Results table 3 is linked to the 1 MeV case.

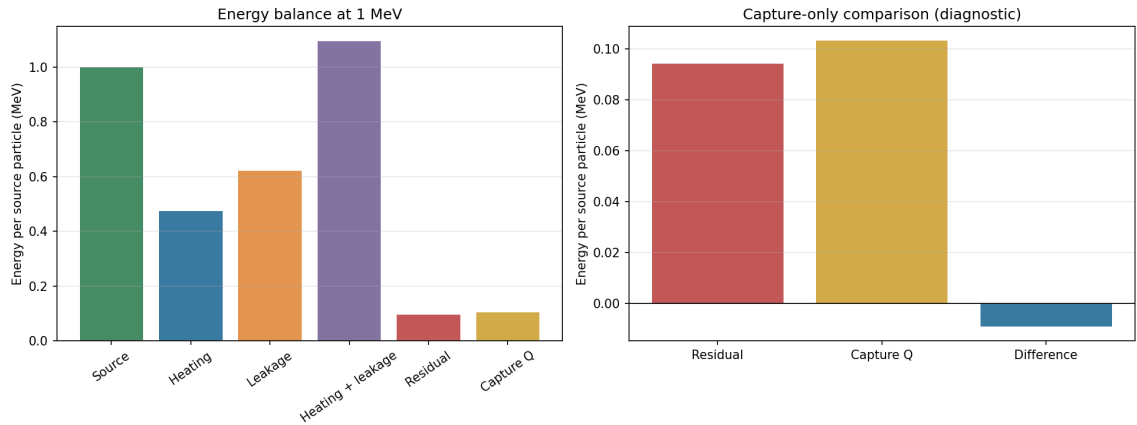


Figure 9: Energy balance with 14 MeV neutron source. The residual is within uncertainty of the caputre reaction energy deposition.

B Quick Install Guide

This quick install guide outlines the basic steps needed to install OpenMC on your computer. For more detailed instructions on configuring and installing OpenMC, see the User's Manual.

Installing on Linux/Mac with Conda

Conda is an open source package management system and environments management system for installing multiple versions of software packages and their dependencies and switching easily between them. OpenMC can be installed in a `conda` environment. First, `conda` should be installed (installation guide) with either Anaconda Distribution or Miniconda. Once you have `conda` installed on your system, OpenMC can be installed via the `conda-forge` channel.

First, add the `conda-forge` channel with:

```
conda config --add channels conda-forge
conda config --set channel_priority strict
```

Then create and activate a new conda environment called `openmc-env` (or whatever you wish) with OpenMC installed.

```
conda create --name openmc-env openmc
conda activate openmc-env
```

If you are installing on macOS with an Apple silicon ARM-based processor, you will also need to specify the `-platform` option:

```
conda create --name openmc-env --platform osx-64 openmc
```

You are now in a conda environment called `openmc-env` that has OpenMC installed.

Installing on Linux/Mac/Windows with Docker

OpenMC can be easily deployed using Docker on any Windows, Mac, or Linux system. With Docker running, execute the following command in the shell to download and run a Docker image with the most recent release of OpenMC from DockerHub:

```
docker run openmc/openmc:latest
```

This will take several minutes to run depending on your internet download speed. The command will place you in an interactive shell running in a Docker container with OpenMC installed.

Note The `docker run` command supports many options for spawning containers including mounting volumes from the host filesystem, which many users will find useful.

Installing from Source using Spack

Spack is a package management tool designed to support multiple versions and configurations of software on a wide variety of platforms and environments. Please follow Spack's setup guide to configure the system.

To install the latest OpenMC with the Python API, use the following command:

```
spack install py-openmc
```

For more information about customizations including MPI, see the detailed installation instructions using Spack. Once installed, environment/lmod modules can be generated or Spack's `load` feature can be used to access the installed packages.

Manually Installing from Source

Obtaining prerequisites on Ubuntu

When building OpenMC from source, all prerequisites can be installed using the package manager:

```
sudo apt install g++ cmake libhdf5-dev libpng-dev
```

After the packages have been installed, follow the instructions to build from source below.

Obtaining prerequisites on macOS

For an OpenMC build with multithreading enabled, a package manager like Homebrew should first be installed. Then, the following packages should be installed, for example in Homebrew via:

```
brew install llvm cmake xtensor hdf5 python libomp libpng
```

The compiler provided by the above LLVM package should be used in place of the one provisioned by XCode, which does not support the multithreading library used by OpenMC. To ensure CMake picks up the correct compiler, make sure that either the `CXX` environment variable is set to the brew-installed `clang++` or that the directory containing it is on your `PATH` environment variable. Common locations for the brew-installed compiler are `/opt/homebrew/opt/llvm/bin` and `/usr/local/opt/llvm/bin`.

After the packages have been installed, follow the instructions to build from source below.

Building Source on Linux or macOS

All OpenMC source code is hosted on GitHub. If you have git, a modern C++ compiler, CMake, and HDF5 installed, you can download and install OpenMC by entering the following commands in a terminal:

```
git clone --recurse-submodules https://github.com/openmc-dev/openmc.git
cd openmc
mkdir build && cd build
cmake ..
make
sudo make install
```

This will build an executable named `openmc` and install it (by default in `/usr/local/bin`). If you do not have administrator privileges, the `cmake` command should specify an installation directory where you have write access, e.g.

```
cmake -DCMAKE_INSTALL_PREFIX=$HOME/.local ..
```

The `openmc` Python package must be installed separately. The easiest way to install it is using `pip`. From the root directory of the OpenMC repository, run:

```
python -m pip install .
```

By default, OpenMC will be built with multithreading support. To build distributed-memory parallel versions of OpenMC using MPI or to configure other options, directions can be found in the detailed installation instructions.

C What's New in 0.15.3

Summary

This release of OpenMC includes many bug fixes, performance improvements, and several notable new features. The major highlights of this release include a new `openmc.deplete.R2SManager` class that automates the workflow for rigorous 2-step (R2S) shutdown dose rate calculations, the ability to collect higher moments for tally results that can be used to test normality, a new uncertainty-aware criticality search method, a new collision tracking feature that enables detailed tracking of particle interactions, support for distributed cell densities, and several new tally filters. The random ray solver also continues to receive significant updates, including automatic setup capabilities, improved geometry handling, and better weight window support. Depletion capabilities have been expanded with thermochemical redox control, external transfer rates, and improved performance.

Compatibility Notes and Deprecations

MCPL has been changed from a build-time dependency to a runtime optional dependency, which means OpenMC will attempt to load the MCPL library at runtime when needed rather than requiring it at build time.

The `openmc.mgxs.Library.add_to_tallies_file` method has been renamed to `openmc.mgxs.Library.add_to_tallies`.

New Features

- A new collision tracking feature enables detailed tracking of particle interactions (#3417)
- Added `openmc.model.Model.keff_search` method for automated criticality searches (#3569)
- Introduced automated workflow for mesh- or cell-based R2S calculations (#3508)
- Ability to source electron/positrons directly for charged particle simulations (#3404)
- Multi-group capability for kinetics parameter calculations with Iterated Fission Probability (#3425)
- Introduced a new `openmc.MeshMaterialFilter` class (#3406)
- Added support for distributed cell densities (#3546)
- Implemented a `openmc.WeightWindowsList` class that enables export to HDF5 (#3456)
- Added `openmc.Material.mean_free_path` method (#3469)
- Introduced `openmc.lib.TemporarySession` context manager (#3475)
- Added material depletion function for tracking individual material depletion (#3420)
- Added methods on `openmc.Material` class for waste disposal rating/classification (#3366, #3376)
- Support for thermochemical redox control transfer rates in depletion (#2783)
- Support for external transfer rates source term in depletion (#3088)
- Added combing capability for fission site sampling and delayed neutron emission time (#2992)
- Ability to specify reference direction for azimuthal angle in `openmc.stats.PolarAzimuthal` distribution (#3582)
- Allow spatial constraints on element sources within `openmc.MeshSource` (#3431)
- Added VTK HDF (.vtkhdf) format support for writing VTK data (#3252)
- Implemented filter weight capability (#3345)
- Optionally collect higher moments for tallies (#3363)

Random Ray Solver Enhancements

- Random Ray AutoMagic Setup for automatic configuration (#3351)

- Point source locator for random ray mode (#3360)
- Support for DAGMC geometries (#3374)
- Optimized mapping of source regions to tallies (#3465)
- Base source region refactor (#3576)

Bug Fixes and Small Changes

- Add two MPI barriers in R2S workflow (#3646)
- Fix a few warnings, rename `add_to_tallies_file` (#3639)
- Fix typo in DAGMC lost particle test (#3634)
- Avoid multiprocessing Pool when running depletion tests with MPI (#3633)
- Support MPI parallelism in R2SManager (#3632)
- Update documentation for particle tracks (#3627)
- Adding variance of variance and normality tests for tally statistics (#3454)
- Avoid divide-by-zero in `from_multigroup_flux` when flux is zero (#3624)
- Write particle states as separate lines in track VTK files (#3628)
- Reset DAGMC history when reviving from source (#3601)
- Add energy group structure: SCALE-999 (#3564)
- Fix bug in normalization of tally results with `no_reduce` (#3619)
- Enable nuclide filters with `get_decay_photon_energy` (#3614)
- Update `check_type` calls to accept both `str` and `os.PathLike` objects (#3618)
- Speed up `apply_time_correction` by reducing file I/O and deepcopies (#3617)
- FW-CADIS Disregard Max Realizations Setting (#3616)
- Random Ray Geometry Debug Mode Fix (#3615)
- Don't write reaction rates in depletion results by default (#3609)
- Allow Path objects in `MGXSLibrary.export_to_hdf5` (#3608)
- Clip mixture distributions based on mean times integral (#3603)
- Allow V0 in `atomic_mass` function (#3607)
- Re-run flaky tests when needed (#3604)
- Ability to load mesh objects from `weight_windows.h5` file (#3598)
- Switch to using coveralls github action for reporting (#3594)
- Add user setting for free gas threshold (#3593)
- Speed up time correction factors (#3592)
- Fix caching issue when using NCrystal materials (#3538)
- Fix random ray source region mesh export (#3579)
- Ensure `weight_windows_file` information is read from XML (#3587)
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- Fix IFP implementation (#3580)
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